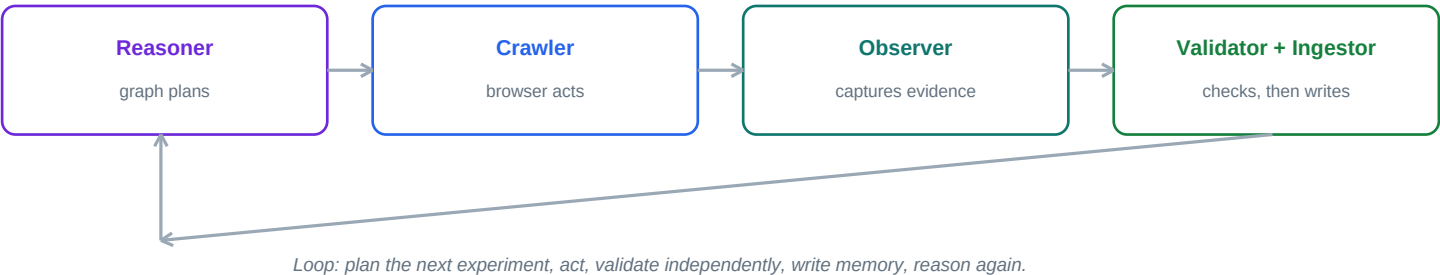


Part B - Production Architecture

Turning the Amazon-checkout prototype into something a real testing platform could run on - the part of the stack a team leans on to decide what is actually safe to ship.

The architecture is built around two things that are painful to change later: the agent harness, where agents live, and the evaluation loop, where you prove the system is improving instead of quietly rotting.



The core loop: the graph plans, the browser acts, the harness validates and writes memory, and the graph reasons over what it learned.

0. The one belief this rests on

A browser agent and a knowledge graph are good at almost opposite things. The browser agent is opportunistic: it can work out which control adds an item to cart after the DOM changes, but it only exercises the paths it happens to walk and can report success on an action that quietly did nothing.

The graph is the part that remembers: every state seen, every control, what was proved versus merely noticed, and what should exist but has not been found. The crawler discovers; the graph remembers and cross-examines; the harness stops the loop from lying to itself.

1. The multi-agent network

The work is split into small agents with hard boundaries rather than one clever mega-agent. Each agent owns one kind of decision, and the hand-off is a typed contract rather than a shared blob of state.

Agent	Owns	LLM?	Boundary enforced
Planner	Frontier, priorities, budgets	No	Decides what to explore; never touches the browser.
Crawler	One UI action via browser-use / Playwright	Small / fast	Executes how; never writes graph truth or decides pass/fail.
Observer	Snapshot: DOM, screenshot, URL, controls	Mostly no	Returns raw evidence, not a trusted summary.
Reasoner	Missed, absent, causal scenarios	LLM proposes; Cypher decides	Deterministic graph queries remain the trustworthy floor.
Validator	Independent post-conditions	No	Nothing is validated until evidence is rechecked.
Healer	Selector repair	Small model + graph	Suggests fixes; Validator confirms.
Triage	Failure clustering and flake calls	LLM + graph history	Escalates; does not silently suppress.
Safety supervisor	Pre-action guardrails	No	Vetoes before the browser sees the task.

Agent	Owns	LLM?	Boundary enforced
Ingestor	Idempotent graph writes	No	Typed, replayable writes only; no model calls.

The most important boundary is between Crawler and Validator. When the Crawler says it changed the quantity, the Validator reads the cart again and checks that the number actually moved.

- **Fails:** retry with backoff up to a budget, then mark the step failed and continue. A failed step is data, not a crash.
- **Stalls:** an action ledger and no-repeat guard turn repeated clicks into a fast veto.
- **Low confidence:** escalate, hand off to a human, or write an explicit unverified marker. Uncertainty is a first-class output.

1.1 The contract between agents

Every hand-off carries intent, expected state, confidence, evidence to seek, risk class, and escalation signal. Those Pydantic-style contracts are the durable interface that lets models or frameworks change underneath.

1.2 How the Validator handles unknown actions

The Validator can only fully confirm an action when it knows what evidence to check. So every action lands in one of three states: done and confirmed, done but unverified, or blocked/failed. The system keeps useful unknown actions, but it does not call them successful until there is proof.

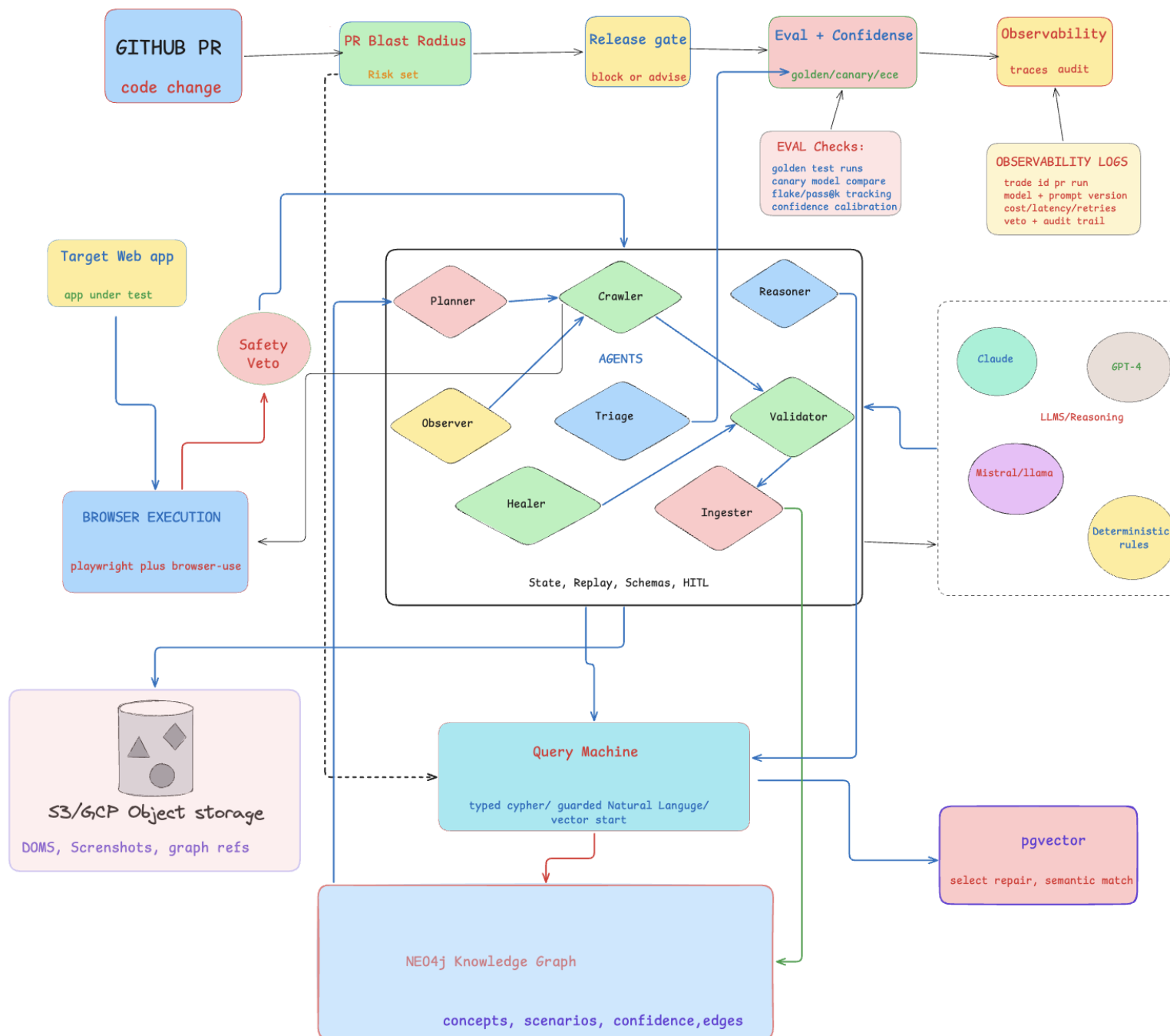
Validator fallback	How it keeps the result honest
Graph expectation	Read cause-and-effect rules such as <code>change_quantity SHOULD_CAUSE subtotal_change</code> .
Always-true invariants	Check facts that should never break, such as totals matching item sums.
Before / after comparison	Look for meaningful state change when an action should have produced one.
Calibrated judge	Use a different model only when no rule or invariant exists, then measure judge agreement against humans.

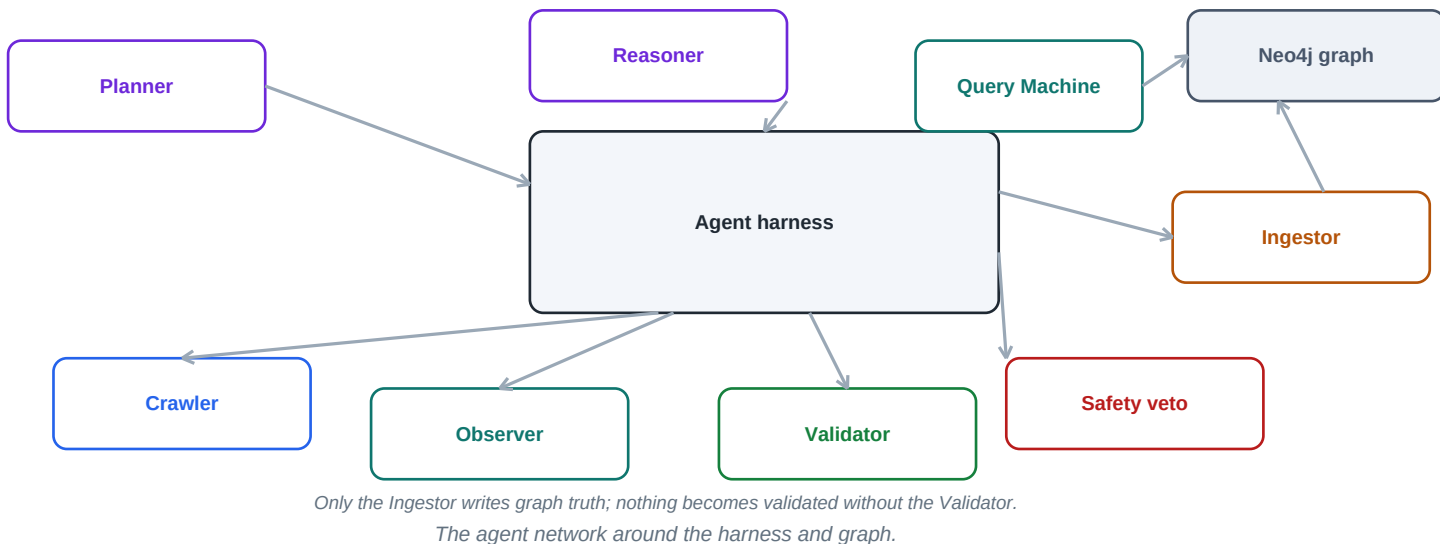
The safe default is simple: admit uncertainty. A tool that tells teams what is safe to ship should prefer 'unverified' over a confident but unsupported pass.

2. System design at a glance

The full production system has three layers: cross-cutting release concerns, the agent runtime that acts on the app, and the memory layer that stores evidence and graph truth.

Hand-drawn system design: release risk, agent harness, browser execution, graph memory, Query Machine, models, and observability.





3. The agent harness - this is the architecture

The harness owns state, retries, timeouts, tool dispatch, schema enforcement, deterministic replay, and human-in-the-loop interrupts. LangGraph can provide state-machine primitives; the durable value sits above it.

- **State and replay:** durable Temporal workflows record inputs, outputs, model and prompt versions, trace IDs, and artifacts.
- **Structured dispatch:** agents return schema-validated objects; malformed output is a caught failure.
- **Retries and budgets:** latency, token, and circuit-breaker policies are observable, not silent spend.
- **HITL interrupts:** runs pause cleanly, ask a human, then resume with provenance intact.

4. Guardrails built into each agent

One shared safety layer across the whole system is the wrong shape because each agent is unsure about different things. Guardrails attach to each agent individually: every agent declares what not confident enough means, and what happens when it crosses that line.

The Part-A deny-list safety veto is the right shape: default-allow reversible exploration, hard-block enumerable irreversible actions such as purchase, payment, sign-out, and navigation away from the product under test.

Guardrail	What it does
Output validators / schema enforcement	The output must fit the contract or the step fails.
Confidence gating	Below threshold: escalate, fall back, or write unverified. No silent low-confidence passes.
Circuit breakers	Repeated failures trip a feature-level stop and surface a flag.
Fallback chains	Frontier model to small model to deterministic rule to human, per agent.
Deny-list veto	Irreversible actions blocked before execution by a separate process.

5. Model routing and composition

The job is choosing the right model for each task and combining them without one model's guess turning into the next one's fact. The principle is deterministic by default, small models for high-volume fuzzy work, and frontier models only where reasoning value is high and volume is low.

Task	Route	Why
State resolution, validation, inference, ingestion	Deterministic code	About 90 percent of operations; must be explainable and repeatable.

Task	Route	Why
Which element to click	Small / fast model	High volume, latency-sensitive UI skill.
Locator stability / flake classification	Fine-tuned small model	Narrow, repetitive, cheaper at scale.
Graph gap reasoning, root-cause chains, eval adjudication	Frontier model	Low volume, high reasoning value.

- Chain models without letting errors snowball: put deterministic checks or graph lookups between model stages.
- Pin prompt and model versions so provider changes are measured before customers file a ticket.
- Keep long sessions small by using the graph as long-term memory and querying only the needed slice.
- Keep frontier models off the hot path; cache small-model calls by graph-state signature and invalidate bounded subgraphs.

Cost shape

The exact provider prices will move, so I would track this with current vendor pricing in production. The architecture target is more important: small models do nearly all live crawl work, graph queries are deterministic, and frontier models run offline or on rare long-tail questions.

Unit	What runs	Budget / cost shape
Per feature crawl	About 25 small-model browser steps, roughly 75k input / 7.5k output tokens	Around cents, not dollars
Graph reasoning	Mostly cached deterministic Cypher; a few small calls near convergence	Near zero on unchanged graph state
Tier-1 graph query	Typed Cypher template	No model cost
Tier-2 graph query	Rare guarded NL-to-Cypher frontier call	Use for under 5 percent of graph questions
50 features nightly	1,500 crawls per customer per month	Keep in the low tens of dollars with caching
Offline frontier work	Rule discovery, eval adjudication, canaries	Amortised across customers

The spend stays controlled because unchanged pages are cached by graph-state signature, PRs invalidate bounded subgraphs instead of the whole app, and the frontier model never sits on the normal crawl path.

6. The production graph schema

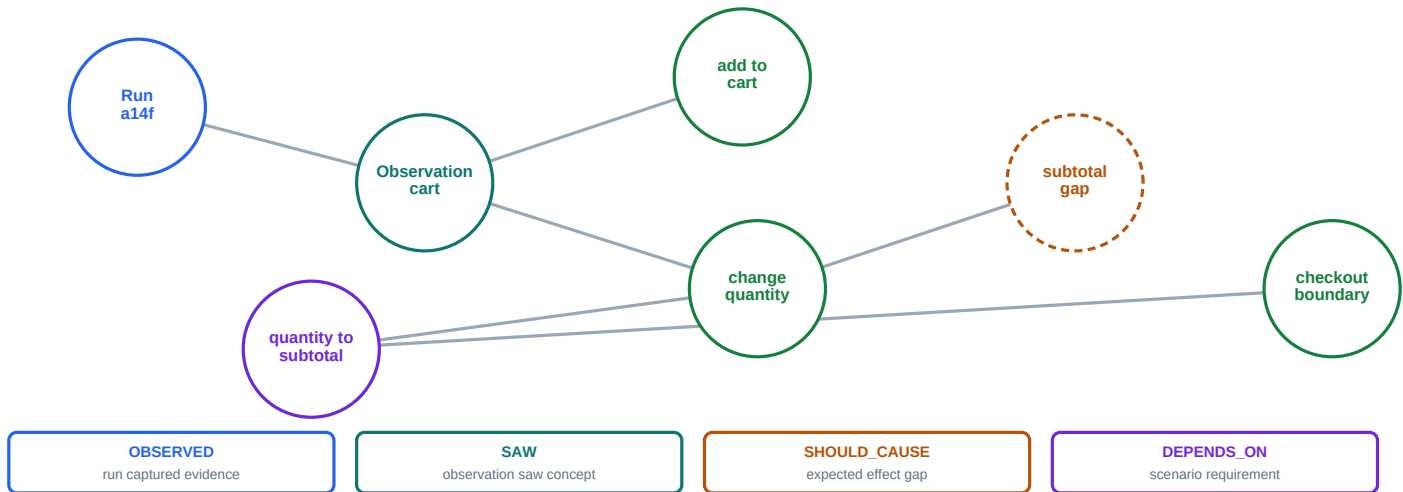
The graph is not built around Element to Component to Flow to Feature. It is built around provenance and absence: why we believe something, and what should exist but does not.

Node	Key properties	Question answered
Tenant / App / Feature	ids	Scope one feature deeply without tenant bleed.
Run	commit_sha, started_at, model_version, cost	What did we know at run N versus N+1?
Observation	state, url, artifact_ref, screenshot_ref	Replay exactly what the browser saw.
Concept	key, kind, observed, validated, expected, confidence	Expected-not-observed; observed-not-validated.
Intent	source, status, risk, selector_ref, confidence	Why an action ran, passed, or was vetoed.
Scenario	key, status, confidence, last_confirmed_run	Which scenarios are trusted now versus decayed?
Selector	hash, role/css/xpath, stability, last_failed	Blast radius and self-healing.
CodeArtifact	path, symbol, commit_sha	Map PR change to selectors to concepts to scenarios.

The unit of meaning is a Concept - a behaviour - not a DOM element. Selectors churn; the meaning of 'add to cart' survives redesigns.

The edges are where most of the value lives, so provenance and confidence belong on edges too: where the relationship came from, and how sure the system is of it.

Edge	Between	What it lets us ask
OBSERVED / SAW_CONCEPT	Run to Observation to Concept	What did we see, and where?
TARGETS	Intent to Concept	Which attempt exercised which behaviour, and did it pass?
DEPENDS_ON	Scenario to Concept	Which concepts does this scenario rely on?
SHOULD_CAUSE	Concept to Concept	Which expected effect remains unproven?
BACKS	CodeArtifact to Selector	Which code backs this locator?



A sample checkout graph: the dashed subtotal node and SHOULD_CAUSE edge are the causal gap the reasoner surfaces.

6.1 Why a graph and not just vectors

- **Absence:** missing tests are structural gaps, not similarity queries.
- **Temporal reasoning:** knowledge at commit X needs stamped edges and history.
- **Multi-hop traversal:** CodeArtifact to Selector to Concept to Scenario is the product.
- Vectors still help with selector repair and semantic matching, but only as an auxiliary index.

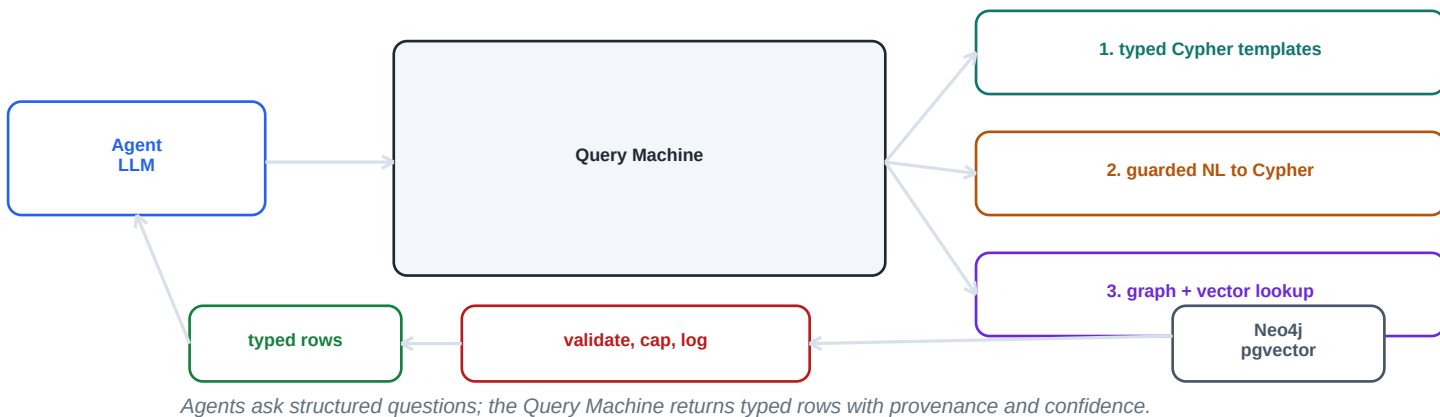
6.2 Indexes and constraints I would declare

A graph schema without indexes is a wish. The indexes should map to the exact queries the platform will run every day.

Index / constraint	Why it exists
Concept(tenant_id, feature_key, key)	Makes concept upserts correct and answers expected-vs-observed questions inside one feature.
Run(tenant_id, app_id, run_id)	Keeps run identity stable and replayable.
Selector(hash)	Fast selector repair and PR blast radius by locator content.
CodeArtifact(commit_sha, path, symbol)	Entry point for the PR hook.
Concept(tenant_id, feature_key)	Hot-path scoped feature lookups.
Run(commit_sha, started_at)	Temporal questions such as what changed between two commits.
SHOULD_CAUSE(valid_from, confidence)	Cheap queries for what the graph believed at time T and how sure it was.

7. The Query Machine

The Query Machine lives between agents and Neo4j. It turns 'what does the graph know?' into bounded, typed answers instead of dumping the graph into a prompt or letting an agent write arbitrary Cypher.



- **Tier 1:** reviewed, parameterised graph-query tools such as `missed_scenarios`, `absence`, `blast_radius`, `neighbours`, `confidence_of`, and `what_did_we_know_at`.
- **Tier 2:** guarded natural-language to Cypher for the long tail: read-only, schema allow-listed, row/cost capped, logged, and reviewable.
- **Tier 3:** hybrid graph plus vector lookup: similarity finds the starting node, structure performs the reasoning.

8. Keeping the graph correct over time

Operation	Trigger	How
Incremental	Every crawl	Upsert observations, bump <code>last_seen</code> , append evidence, add <code>Scenario-[CONFIRMED_BY]->Run</code> .
Recomputed	A run touching a feature	Re-run inference and confidence for that feature only; never global.
Invalidated	PR touches mapped code; selector fails repeatedly; validation fails	Mark a bounded subgraph stale and push affected scenarios into the retest frontier.
Compounds	Many runs over time	Selector stability, flake rate, transition probabilities, and confidence-decay curves.

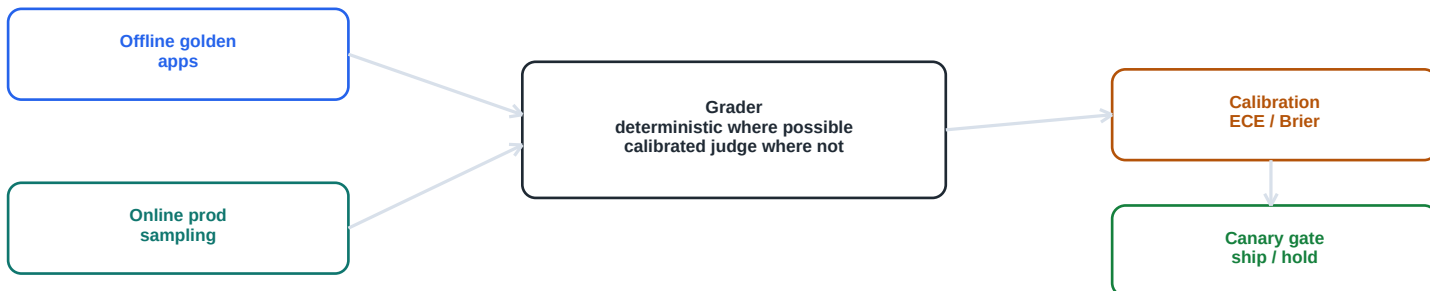
- Conflict resolution preserves history instead of overwriting: old and new observations remain stamped by commit and time, so what the system believed at commit X stays answerable.
- Validated decisions feed the next run: confirmed scenarios raise priors; repeatedly flaky ones lower them.
- Full nightly recompute is rejected as the freshness story. Bounded invalidation is the defensible production shape.



Confidence decays when runs do not reconfirm a scenario and snaps back on proof. Below the retest line it re-enters the frontier.

9. Eval and confidence

This is the layer that decides whether the system is a product or a science project. It protects against two confidently-wrong states: the agent clicked the wrong thing, and the graph inferred a scenario that is not real.



Offline golden sets and online sampling feed graders; calibration and canary gates stand between a model change and customers.

Question	Signal	Metric
Did it click the right target?	Trace plus positive/negative target check	Wrong-target rate, wander rate
Did the action succeed?	Deterministic post-condition after re-observe	Validation pass rate
Is the scenario reproducible?	Replay across clean sessions	Flake rate, pass@k
Are confidence scores honest?	Predicted confidence versus observed reproduction	ECE, Brier score
Did a model change regress us?	Canary on versioned golden trajectories	Delta versus baseline

Use a managed eval store to move fast, but own the golden sets and grading logic in-house. LLM-as-judge must be calibrated against human spot-checks, and never used where a deterministic check exists.

10. Observability and operations

- One trace id follows the whole chain of agent calls, with model and prompt versions stamped at each hop.
- Decision audit trails record frontier score, guardrail choice, confidence, and veto/pass outcome.
- Latency and cost budgets per stage make stalls visible.
- Alerts focus on agentic failure modes: wander-rate drift, validation spikes, stale concepts, and model-upgrade regressions.
- Rollout moves from observe-only to recommend to gated execution to autonomous safe execution after thresholds clear.

11. Multi-tenancy and scale

Layer	Isolated per tenant	Shared
Graph	Tenant/app/feature partition; database-per-tenant for large accounts	Schema and inference engine code
Artifacts	Per-tenant namespace, retention, redaction	Storage implementation
Models / prompts	Customer vocabulary and trace versions	Generic templates
Learning	No selectors, DOM, screenshots, or scenarios cross tenants	Aggregated, reviewed structural priors only

Scale shape: async agents on a queue, durable Temporal workflows, LLM-call caching by graph-state signature, bounded Query Machine traversals, and heavy DOM/screenshot artifacts in object storage while Neo4j stores references.

12. PR blast radius

The PR hook falls straight out of the graph: CodeArtifact to Selector to Concept to Scenario to confidence. The output should say exactly which selectors, validated scenarios, and inferred scenarios are at risk - not 'rerun all checkout tests.'

13. Production concerns beyond the checklist

Concern	Production stance
Security and privacy	DOM, screenshots, and sessions are sensitive. Encrypt per tenant, redact PII/tokens before storage, cap screenshot retention, and never log cookies or auth tokens.
Test data and safe environments	Use dedicated test accounts and seeded staging/sandbox data. The system can prove it reached a checkout boundary; it must not cross into real purchase.
Auth and sessions	A human handles OTP/CAPTCHA once; encrypted per-tenant sessions are refreshed safely. Account-level controls stay on the deny-list.
CI and release gates	Start in recommend mode, then earn blocking gates only after confidence and flake metrics hold. Known flaky tests should not block good releases blindly.

14. What I would refuse to ship

Refusal	Why, and what must exist first
A path that can click final purchase	Money pages must be structurally incapable of final submit; red-team it until it cannot buy.
LLM self-report as validation	Every mutating action needs a deterministic post-condition.
Uncalibrated confidence in the UI	Confidence without calibration is false authority.
Nightly full-graph recompute as freshness	Expensive and hides drift; prove bounded invalidation on a golden tenant.
Unreviewed cross-tenant learning	Only reviewed structural priors may cross tenants; never DOM, selectors, screenshots, or scenarios.
Hard-coded concept vocabulary as-is	Fine for one feature; production needs a learned per-tenant concept model behind an adapter.

The hardest problem is knowing the system is wrong before a human notices. The architecture answers with deterministic floors under the fuzzy parts, confidence that is measured rather than assumed, canaries between model changes and customers, and a graph that remembers what used to be true so drift is visible.

15. How Part A grounds every claim

Production claim	Prototype seed
Planner / Crawler split; graph plans, browser acts	GraphGuidedExplorer._autonomous_loop; BrowserUseIntentExecutor
Guardrails an LLM cannot bypass	safety_guard.veto_reason in the browser-use callback
Executor output is evidence, not truth	re-observe plus _run_assertions; cart-delta / checkout verification
Graph reasons about missed and causal gaps	infer_missed_scenarios; CAUSAL_EXPECTATIONS / SHOULD_CAUSE
Query Machine tier 1 and NL taste	infer_missed_scenarios; blast_radius; graph_expansion.expand_from_graph
Absence as graph query	seed_expected_concepts; missing_expected_concepts
PR blast radius traversal	pr_blast_radius.py; GraphStore.blast_radius
Provenance and confidence on writes	write_intent; write_observation

A concrete run makes the point clearer.

- **Discovered by the crawler:** Add to Cart works, and the quantity stepper moves. The Validator confirmed this by re-reading the actual cart.

- **Inferred by the graph:** Proceeding to checkout should reach the checkout boundary. The free crawl saw the control but never proved the effect.
- **Why the graph inferred it:** `action.proceed_to_checkout SHOULD_CAUSE domain.checkout_boundary`. Observed cause, expected effect, no proof.
- **What happened next:** the graph turned that gap into a directed probe, the browser clicked Proceed, and the run reached the real checkout boundary.

16. The stack, and why

Layer	Start with	Avoid
Agent harness	Custom harness over LangGraph primitives; Temporal for durable runs	A fully autonomous agent with no state/replay contract
Browser execution	Playwright plus browser-use, sandboxed	No action contract or veto
Graph store	Neo4j with temporal provenance and confidence on edges	Graph-shaped dependencies hidden in relational tables
Retrieval	pgvector / Weaviate as auxiliary hybrid lookup	Vector similarity replacing graph truth
Query Machine	Parameterised Cypher tools plus guarded NL-to-Cypher	Free-form model-written Cypher on the hot path
Models	Frontier reasoning/judge; small executors; fine-tuned narrow classifiers; deterministic validation	Provider name as the whole strategy
Eval	Versioned golden apps, managed eval store, calibrated judge, canaries	Shipping prompt changes without canaries
Observability	OpenTelemetry, trace IDs, decision audit	Plain logs with no trace IDs
Async / queue	Celery + Redis, FastAPI, Docker/K8s, AWS/GCP	Fire-and-forget cron for retryable crawls
Artifacts	S3/GCS; Neo4j holds refs only	Large DOM/screenshot blobs inside Neo4j

In one line: the valuable part is not the clever crawler. It is the loop between something that explores and something that remembers, with a hard wall between what the system did and what it is allowed to believe.